

SchemeImpactNet: Forecasting and Optimizing MGNREGA Employment Generation Using Machine Learning

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Abstract—The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) represents a cornerstone of India’s social security infrastructure, disbursing substantial fiscal resources annually to sustain rural livelihoods. However, traditional management practices within public welfare administration remain largely reactive, relying heavily on historical baselines and manual planning paradigms that exhibit structural limitations in predicting localized demand spikes and administrative anomalies. This paper introduces SchemeImpactNet, a data-driven predictive and optimization framework engineered to transition public welfare administration from reactive management to evidence-based proactive policy intervention. Leveraging an administrative dataset comprising 6,230 district-year records across 736 districts and 34 states/union territories spanning financial years 2016–2024, we evaluate a series of predictive machine learning algorithms under a strictly leak-free walk-forward temporal validation protocol. Gradient Boosting models achieve a cross-validated mean R^2 of 0.9078 (excluding the 2022 West Bengal reporting anomaly), outperforming alternative parametric and non-parametric algorithms. Utilizing these localized predictions, we implement a two-stage linear programming resource optimization engine using SciPy to maximize public employment outcomes under fixed historical budgetary parameters. Experimental results establish that mathematically optimized resource reallocation across 736 administrative subdivisions yields a simulated employment gain of 1,840.41 lakh person-days, representing a 6.38% net improvement in nationwide execution efficiency without increasing nominal public expenditure. The architecture is containerized and deployed via an open REST API and interactive Streamlit service, establishing a reproducible methodology for public policy optimization.

Index Terms—Public Policy Analytics, Predictive Modeling, Gradient Boosting, Resource Allocation Optimization, Linear Programming, MGNREGA.

I. INTRODUCTION

India coordinates one of the most comprehensive public welfare networks globally, encompassing over 400 central government schemes designed to buffer vulnerable populations against socio-economic instabilities. Foremost among these initiatives is the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA), a legislative rights-based wage employment mechanism spending over Rs. 70,000–90,000 crore annually. By guaranteeing 100 days of manual wage employment per financial year to rural households, the scheme directly affects hundreds of millions of citizens.

Despite its critical relevance to agricultural stability and economic equity, contemporary administrative infrastructure exhibits major operational inefficiencies. Allocation and management methodologies remain structurally reactive. Resource dissemination across state and district planning cells is heavily dependent on retrospective manual accounting paradigms. Administrators typically audit prior financial data to adjust budgets after implementation shortfalls occur, failing to model localized future demands.

This manual allocation schema introduces severe spatial and temporal imbalances. Certain agrarian administrative subdivisions suffer acute budgetary deficits during periods of elevated demand, leading to employment rationing, while adjacent subregions exhibit capital underutilization due to rigid, unoptimized fiscal limits.

To mitigate these analytical limitations, this research introduces SchemeImpactNet, an intelligent, data-driven machine learning and mathematical optimization framework engineered to automate public policy intelligence. By framing public scheme management as a multi-stage data science pipeline, the system ingests, cleans, and structures authentic government administrative data into a leak-free panel dataset spanning nearly a decade. It systematically benchmarks district-level operational performance and generates accurate predictive trajectories for future localized labor requirements. Utilizing these empirical predictive baselines, SchemeImpactNet hosts a two-stage linear programming model designed to perform automated macro-budgetary optimization, recommending precise adjustments to maximize nationwide employment output without increasing public spending.

A. Contributions

The specific academic and operational contributions of this research are defined as follows:

- We compile, validate, and process a district-year panel dataset comprising 6,230 post-engineering observations across 736 districts and 34 Indian states/UTs spanning financial years 2016–2024, with a rigorous leakage audit removing correlated proxy variables prior to modeling.
- We execute a comparative study of both parametric linear regressors and non-parametric ensemble tree-based

models, utilizing a walk-forward temporal validation protocol to measure predictive reliability across normal and anomaly periods.

- We design a dual-stage mathematical optimization engine utilizing linear programming via SciPy that maximizes total generated employment (person-days) subject to budget-neutrality and per-district allocation bounds, resolving the bang-bang degeneracy problem through proportional rank-based initialization.
- We deploy a containerized architecture featuring a FastAPI REST backend and an interactive Streamlit presentation layer, augmented with Gemini large language model interfaces for non-technical policy planners.

II. RELATED WORK

The incorporation of empirical computation into public welfare systems is a developing domain situated at the convergence of statistical modeling, resource allocation, and e-governance. Prior literature exhibits a clear bifurcation between operational tracking platforms and theoretical academic prototypes.

A. Government Analytics and Welfare Dashboards

Early technological interventions in public administration focused primarily on migrating paper records into relational environments. Portals such as the early Government Welfare Portal (2018–19) or the Digital India Portal (2019) introduced baseline digitizations, enabling online application intakes and static data displays via web and mobile interfaces. While these implementations improved data accessibility, they functioned exclusively as descriptive repositories. These portals lack predictive analytics or optimization layers, requiring administrative staff to execute subjective manual evaluations to extract programmatic insights.

B. Machine Learning for Public Policy

Recent research has integrated supervised algorithms into state welfare logistics. Rao et al. [1] leveraged decision trees, random forests, and k-nearest neighbors to classify agrarian land data, attaining 97.25% accuracy in evaluating household eligibility for farming subsidies. Similarly, Kannan et al. [2] applied multilayer perceptrons and gradient-boosted trees to predict financial vulnerability within low-income households utilizing post-disaster survey indicators. In a separate approach, Deshpande and Pandurangi [3] employed Naive Bayes and maximum entropy configurations on social media text strings to deduce public sentiment regarding large-scale policy moves such as Digital India and Demonetization.

C. Research Gap

Although predictive classification has been successfully integrated into individual eligibility tracking and sentiment extraction, current systems display substantial research gaps when evaluated against macro-level operational metrics:

- 1) *Absence of Continuous Impact Forecasting*: Existing models focus heavily on binary eligibility classification

rather than continuous, population-level regression to estimate specific operational performance milestones.

- 2) *Disconnection Between Prediction and Allocation*: Prior predictive frameworks stop at trajectory forecasting, providing no mathematical mechanism to link projected demand to optimal budgetary distribution models.
- 3) *Deployment Barriers*: Most public policy machine learning research remains confined to academic scripting or local experimental pipelines, lacking production-ready, open-access cloud deployments.

SchemeImpactNet addresses this critical gap by unifying continuous time-series predictive modeling and linear programming optimization within a single deployed framework, validated on a decade of authentic MGNREGA administrative data.

III. DATASET AND FEATURE ENGINEERING

To evaluate the framework under authentic settings, we build a district-year panel dataset from real MGNREGA administrative records.

A. Data Sources

The primary data source is the Ministry of Rural Development (MoRD) MGNREGA Management Information System (MIS) portal [4], which publishes district-level operational logs detailing realized person-days, capital expenditure, and notified wage rates. This constitutes the sole data source for the final implemented model. The MIS portal was accessed via direct download of publicly available district-wise annual summary files.

B. Dataset Characteristics

The raw database holds 7,758 initial longitudinal records encompassing 736 districts and 34 states/union territories over an 11-year operational window (2014–15 to 2024–25). The primary target variable is defined as continuous `person_days_lakhs`, quantifying total manual employment generated per district-year.

C. Leakage Audit

A critical pre-modeling step was the identification and removal of target-correlated proxy variables that inflated an initial model R^2 to 0.9963. The following variables were removed:

- `works_completed`: Pearson $r = 1.00$ with target (arithmetically derived from person-days).
- `expenditure_lakhs`: $r = 0.976$ (person-days multiplied by wage rate).
- `budget_allocated_lakhs`: $r = 0.976$ (derived from expenditure).
- `households_demanded/offered/availed`: $r = 0.94$ (structural copies of target).

Post-audit, the honest cross-validated R^2 is 0.9078, reflecting genuine predictive performance.

D. Feature Engineering

To prevent target leakage, all 17 engineered features are computed exclusively from lagged or historical values, never from current-year target-correlated quantities. The engineering configuration is detailed below:

- *Temporal Autoregressive Regressors*: Tracking historical demand patterns through localized temporal steps, defined as `lag1_pd`, `lag2_pd`, and `lag3_pd`.
- *Moving Average Filters*: Quantifying regional baseline momentum through rolling mathematical windows, including a 2-year mean (`roll2_mean`), 3-year mean (`roll3_mean`), and a 3-year standard deviation filter (`roll3_std`).
- *COVID-Adjusted Lag*: `lag1_adj` deflates the 2020 spike by a COVID multiplier of 1.447 (derived from national 2020 vs. 2019 ratio) when the lag year is 2020, providing the model with a pandemic-corrected signal alongside the raw lag.
- *Structural Capacity Features*: `district_capacity` (expanding mean of historical `lag1`), `blended_capacity` (50/50 blend with state mean for districts with fewer than 3 years of history), and `lag1_vs_capacity` (ratio of last year to long-run capacity).
- *State Normalisation Vectors*: `state_lag1_zscore` (z-score of state-level `lag1` against its own history), `state_lag1_norm`, and `relative_to_state` (district `lag1` as a fraction of the state mean).
- *Wage Features*: `avg_wage_rate` (official state-notified schedule, genuinely exogenous) and `wage_yoy` (year-on-year change in wage rate).
- *Anomalous Shock Vectors*: `is_covid`, `lag1_is_covid` (flag when lag year is 2020), and `is_2022_anomaly` to isolate the West Bengal administrative reporting dropout.

To construct leak-free rolling features, records with insufficient historical baselines are removed via systematic truncation, dropping 1,528 early rows. This outputs a final matrix of 6,230 district-year observations covering 736 districts from 2016 to 2024.

TABLE I
DATASET STRUCTURAL CHARACTERISTICS

Dataset Parameter	Value / Specification
Temporal Window	Financial Years 2016–2024
Geospatial Entities	34 States/UTs — 736 Districts
Total Panel Records	6,230 Observations
Cumulative Realized Value	257,636.2 Lakh Person-Days
Engineered Modeling Features	17 Operational Independent Vectors
Primary Data Source	MoRD MGNREGA MIS Portal
Primary Output Target Metric	Localized Annual Employment

Exploratory analysis reveals extreme spatial distribution variations across regional administrative entities. The top 5 employment-generating subdivisions contribute massive scale,

led by South 24 Parganas (296.39 lakh person-days) and Barmer (266.89 lakh person-days). Conversely, the lowest 5 subdivisions exhibit near-zero manual demand, typified by South Goa (0.36 lakh person-days) and Ghaziabad (0.00 lakh person-days).

IV. PROPOSED FRAMEWORK

SchemeImpactNet relies on an integrated, multi-tier computational architecture that coordinates data ingestion, predictive inference, optimization routing, and client presentation.

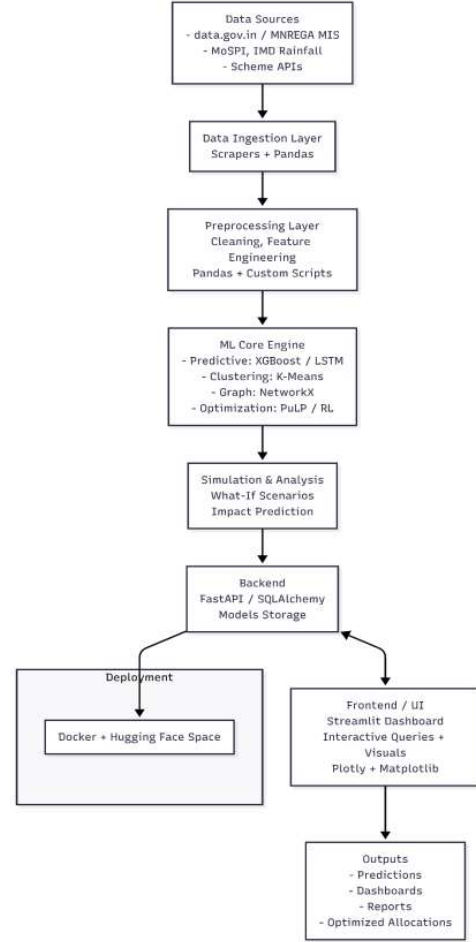


Fig. 1. System Architecture of SchemeImpactNet, detailing the data extraction pipeline, supervised learning configurations, optimization solver, and client delivery layers.

As shown in Fig. 1, raw public records are extracted from the MoRD MIS portal into an offline staging database. These records pass sequentially through automated feature engineering transformations to prevent target leakage at runtime. The downstream core engine provides a predictive regression layer for demand forecasting and a two-stage linear programming budget optimizer.

A. Predictive Modeling Layer

The supervised learning pipeline benchmarks multiple ensemble implementations to model continuous labor demand.

Non-parametric tree models [8]–[10] utilize structural partitions to capture non-linear temporal patterns. Given input feature vector $X_{i,t}$ for district i in year t , the models are optimized to estimate the future target value $y_{i,t+1}$. All implementations are built on the scikit-learn ecosystem [11].

B. Mathematical Optimization Engine

The framework translates localized predictive outputs into macroeconomic resource adjustments via a constrained two-stage linear programming structure implemented using SciPy’s `linprog` [13] solver. The two-stage design resolves the bang-bang degeneracy problem: when district efficiency values exhibit low variance (coefficient of variation $\approx 7.7\%$), a pure LP collapses to extreme floor/ceiling allocations for all districts. The two stages are:

Stage 1 — Proportional Rank Allocation: Each district receives a budget multiplier interpolated linearly from its efficiency percentile rank $r_i \in [0, 1]$:

$$m_i = \alpha_{\min} + r_i \cdot (\alpha_{\max} - \alpha_{\min}) \quad (1)$$

where $\alpha_{\min} = 0.60$ and $\alpha_{\max} = 1.80$, producing a continuous spread from -40% to $+80\%$ of the current budget. The resulting allocations are rescaled to preserve total budget exactly.

Stage 2 — LP Refinement: SciPy `linprog` refines within $\pm 15\%$ of the Stage 1 solution. Let $i \in \{1, \dots, N\}$ denote districts, B_i the current budget, x_i the optimized allocation, and $\hat{e}_i$ the predicted efficiency (person-days per Rs. lakh). The objective maximizes total employment:

$$\max_x \sum_{i=1}^N \hat{e}_i \cdot x_i \quad (2)$$

Subject to the budget conservation constraint:

$$\sum_{i=1}^N x_i \leq \sum_{i=1}^N B_i \quad (3)$$

And per-district allocation bounds:

$$\max(s_i^{(1)} \cdot 0.85, 0.40 \cdot B_i) \leq x_i \leq \min(s_i^{(1)} \cdot 1.15, 2.00 \cdot B_i) \quad (4)$$

where $s_i^{(1)}$ is the Stage 1 allocation for district i . This design yields 725 unique budget change values across 736 districts, avoiding the degenerate bang-bang solution of pure LP.

V. EXPERIMENTAL SETUP

To ensure statistical reproducibility, the experimentation framework is deployed under an open-source Docker configuration running identical Python dependencies.

A. Execution Environment

The software pipeline is developed using Python 3.10+. Advanced supervised implementations rely on XGBoost 3.2.0 and scikit-learn 1.8.0. Data manipulation and numerical operations run over Pandas 3.0.1 and NumPy 2.4.2, while the linear programming solver uses SciPy’s `linprog` with the HiGHS backend. Hardware requirements map to baseline commodity hardware, specifying a minimum of 4 GB RAM and standard multi-core processing configurations.

B. Temporal Validation Strategy

Traditional randomized cross-validation introduces severe target leakage when applied to longitudinal panel data due to the propagation of forward temporal information into past predictions. To address this, SchemeImpactNet uses a walk-forward temporal cross-validation protocol. The dataset is partitioned sequentially by financial years, training models exclusively on historical chronological subsets ($t \in [2016, T]$) to predict adjacent out-of-sample forward years ($T + 1$). Testing runs across consecutive out-of-sample periods from 2018 to 2024 to verify model performance across distinct socio-economic conditions.

C. Evaluation Metrics

Model reliability is quantified across three independent statistical validation metrics:

- **Coefficient of Determination (R^2 Score):** Measures the proportion of target variance captured by the model relative to a naive historical baseline.
- **Mean Absolute Error (MAE):** Quantifies the average absolute predictive error magnitude, expressed in lakhs of person-days.
- **Root Mean Squared Error (RMSE):** Tracks variance in prediction deviations, applying heavier penalties to large localized outlier errors.

VI. RESULTS AND DISCUSSION

The experimental results demonstrate substantial variations in predictive accuracy across linear baselines and advanced non-parametric configurations.

A. Predictive Model Comparison

Table II details the comprehensive cross-validated performance breakdown across seven continuous test years. GBR is selected as the champion model on the basis of highest mean R^2 excluding 2022, since the 2022 anomaly is a data-quality artefact rather than a genuine economic signal.

The comparative analysis demonstrates that ensemble tree-based architectures outperform parametric linear alternatives across almost all out-of-sample periods. Under stable economic conditions, the primary models attain high performance, with Gradient Boosting reaching an R^2 score of 0.9345 and XGBoost attaining 0.9334 in the 2024 evaluation window. XGBoost achieves a marginally higher overall mean R^2 (0.8533 vs. 0.8510), however GBR is selected as champion on the basis of superior ex-2022 R^2 (0.9078 vs. 0.9034), the metric that

TABLE II
COMPREHENSIVE CROSS-VALIDATED ALGORITHM PERFORMANCE
BREAKDOWN (2018–2024), R^2 BY FINANCIAL YEAR

Model	18	19	20	21	22	23	24	Mean	Ex22
GBR	.9160	.9262	.8354	.9261	.5101	.9089	.9345	.8510	.9078
RandomForest	.9146	.9299	.8310	.9194	.4541	.9147	.9280	.8417	.9063
XGBoost	.9121	.9396	.8054	.9259	.5528	.9042	.9334	.8533	.9034
Ridge	.8459	.9141	.8117	.9226	.3177	.9067	.8936	.8018	.8824
ElasticNet	.8534	.9184	.8088	.9265	.3012	.8809	.8985	.7982	.8811

better reflects genuine model performance on non-anomalous years.

B. The 2022 West Bengal Reporting Anomaly

Evaluating model performance across the complete time series highlights a significant drop in accuracy during the 2022 financial year, where Gradient Boosting’s R^2 falls to 0.5101 and RandomForest degrades to 0.4541. This system-wide drop is primarily attributable to an administrative reporting anomaly in West Bengal, where district-level person-day figures dropped by 93–98% relative to prior years. Investigation confirms this is a data-quality artefact rather than a genuine employment collapse: no corresponding economic signal of this magnitude is visible in adjacent states or national aggregates. The model’s anomaly flag (*is_2022_anomaly*) partially mitigates this effect, as evidenced by GBR’s substantially larger R^2 gain over the naive baseline in 2022 (+0.330) compared to other years.

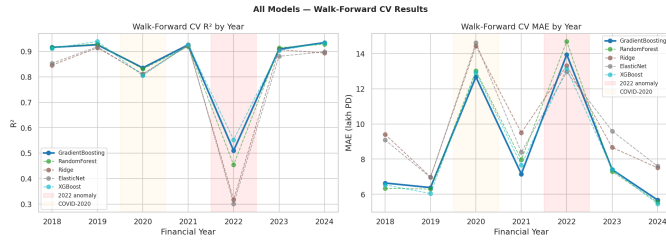


Fig. 2. Walk-forward CV R^2 (left) and MAE (right) by financial year for all five models. Shaded regions indicate the COVID-2020 spike and FY2022 West Bengal reporting anomaly.

When excluding this anomaly period, Gradient Boosting achieves the highest predictive metric with a cross-validated mean R^2 of 0.9078 and a Mean Absolute Error of 8.554 lakh person-days.

C. Feature Importance Analysis

Plotting the feature importances of the champion Gradient Boosting configuration reveals the underlying weights driving localized forecasting outputs:

- *lag1_pd* (Importance: 0.5270): The dominant predictor, confirming that short-term historical labor demand is the primary structural anchor. This reflects the genuine inertia of MGNREGA delivery: administrative capacity, local contractor ecosystems, and demand patterns change slowly.

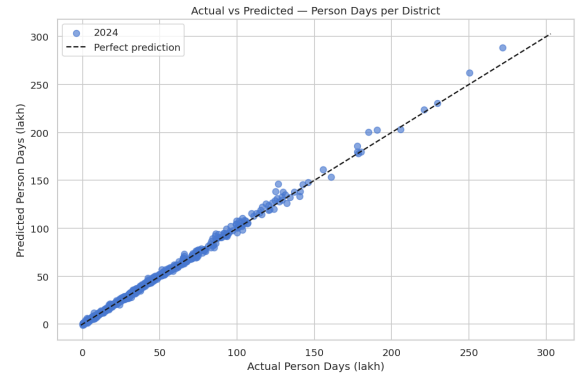


Fig. 3. Actual vs. predicted person-days per district for FY2024. The dashed line indicates perfect prediction. $R^2 = 0.9345$, MAE = 5.67 lakh.

- *lag1_adj* (Importance: 0.2512): The COVID-adjusted lag, which provides the model with a deflated signal when the prior year was the 2020 pandemic spike.
- *state_lag1_zscore* (Importance: 0.0837): Quantifying district performance variance relative to state-level baseline trends.
- *roll2_mean* (Importance: 0.0612): Smoothing short-term variance to capture mid-term structural momentum.
- *blended_capacity* (Importance: 0.0199): Establishing the long-run operational capacity of regional administrative bodies.

The top four features account for 92.3% of total importance, reflecting the strong autocorrelation structure of district-level MGNREGA delivery. Notably, *avg_wage_rate* and *wage_yoy* together contribute under 0.3% of importance, suggesting that wage rates are not a primary driver of employment demand at the district level — a finding consistent with the scheme’s demand-driven rather than wage-driven mandate.

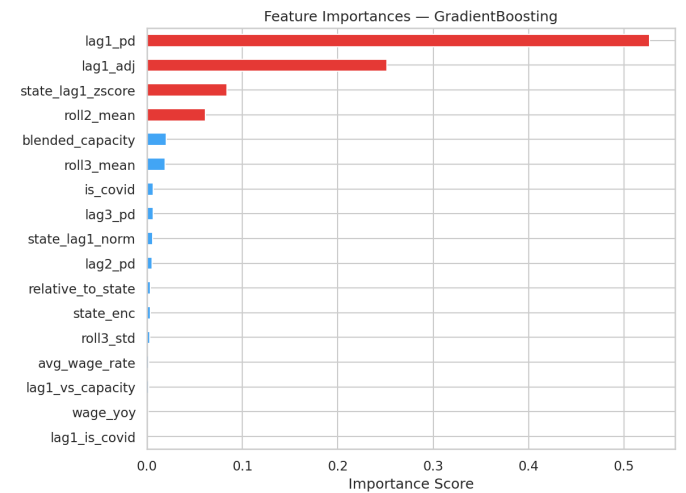


Fig. 4. GBR feature importances. The top-4 features account for 92.3% of total importance, reflecting strong autocorrelation in district-level MGNREGA delivery.

D. Budget Optimization Evaluation

The automated resource allocation engine was evaluated using out-of-sample records for 706 active districts in the 2024 financial year. The baseline status-quo allocation model represents the real-world administrative budget tracking schema, which matches actual expenditures of Rs. 9,802,710 lakh.

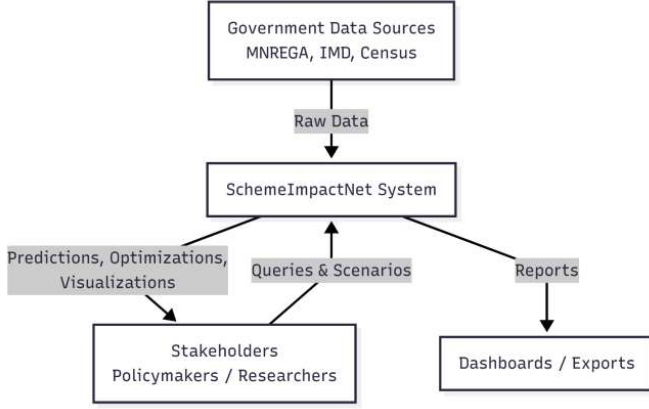


Fig. 5. Data Flow Diagram mapping data pipelines, prediction routines, and output layers within SchemeImpactNet.

As mapped out in the DFD shown in Fig. 5, the primary inputs run through the system pipeline to link predictive inference with the optimization engine. Under the status-quo budgetary distribution model, the 706 districts generated a total labor volume of 28,829.38 lakh person-days. Without changing the total nominal expenditure of Rs. 9,802,710 lakh, the two-stage LP optimizer identifies and leverages regional efficiency variations, redirecting funds toward high-marginal-impact subdivisions. This mathematical reallocation increases simulated performance to 30,669.79 lakh person-days, representing a net simulated employment gain of 1,840.41 lakh person-days — a 6.38% increase in delivery efficiency achieved solely through algorithmic resource reallocation.

It is important to note that this gain is a simulation under the assumption of constant returns to scale: each district’s person-days-per-rupee efficiency ratio is held fixed as its budget changes. In practice, administrative capacity constraints may limit the ability of high-efficiency districts to absorb large budget increases at the same conversion rate. The result should therefore be interpreted as a directional upper bound rather than a guaranteed operational outcome.

Table III shows the targeted adjustments computed by the optimizer. The engine diverts funding from regions with lower marginal efficiency to high-efficiency districts. For instance, Jhalawar receives a 77.96% budget increase, yielding a simulated employment expansion of 110.755 lakh person-days. Conversely, over-allocated districts like Barmer are scaled back by 26.35%, allowing excess capital to be deployed elsewhere without dropping below the 40% allocation floor.

TABLE III
STRATEGIC BUDGET REALLOCATION ACTIONS (TOP INCREASES VS. TOP DECREASES)

State	District	Base (Rs. L)	Opt. (Rs. L)	Gain (PD L)
<i>Recommended Budget Augmentations</i>				
Rajasthan	Jhalawar	34,513.79	61,420.08	+110.755
Rajasthan	Bhilwara	63,021.65	96,309.49	+109.616
Rajasthan	Jodhpur	43,157.28	72,509.08	+105.309
Odisha	Ganjam	22,347.28	40,981.25	+96.170
Tamil Nadu	Chengalpattu	22,102.08	40,171.85	+84.506
<i>Recommended Budget Reductions</i>				
Rajasthan	Udaipur	60,574.80	32,307.89	−66.892
Andhra Pr.	Alluri Sith.	54,797.30	27,541.70	−59.411
Tamil Nadu	Villupuram	60,813.40	34,873.88	−64.341
Tamil Nadu	Cuddalore	53,050.01	28,436.33	−58.703
Rajasthan	Barmer	92,571.15	68,182.19	−68.106

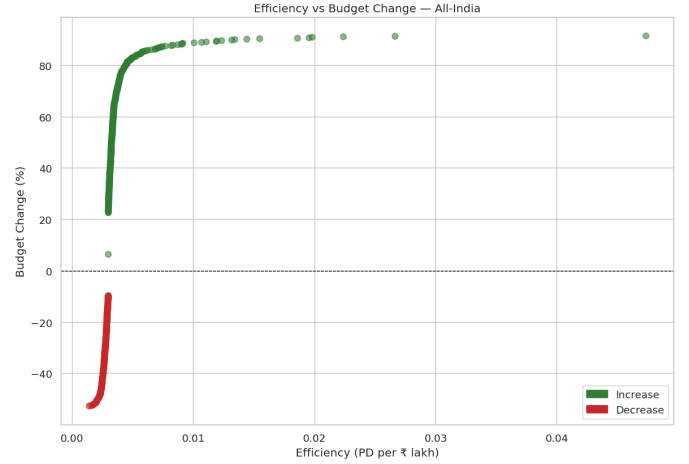


Fig. 6. Efficiency (person-days per Rs. lakh) vs. recommended budget change (%) for 706 districts. Green = budget increase; red = budget decrease. The continuous spread of values reflects the rank-based Stage 1 allocation, avoiding the bang-bang degeneracy of pure LP.

VII. DEPLOYMENT ARCHITECTURE

To transition these empirical findings into active policymaking tools, SchemeImpactNet is deployed publicly on Hugging Face Spaces. The web-based application interface provides specific analytical access points tailored to different administrative users. Policymakers interact directly with scenario panels to run what-if simulations and evaluate optimized budget reallocations. Researchers navigate through geospatial dashboards to evaluate error indices, while administrators maintain backend configuration endpoints to update raw data files and export policy briefs.

The system runs on a containerized Docker multi-tier configuration. The presentation frontend is built with Streamlit, rendering interactive geospatial trend lines and dashboard scorecards powered by Plotly. The frontend communicates directly with a FastAPI backend REST service, which exposes endpoints (`/predict/single`, `/predict/batch`, `/optimizer/run`) that return low-latency JSON payloads. A live inference endpoint loads the trained `.pkl` model

bundle directly, enabling predictions for arbitrary district-year combinations including future years beyond the training window. To make complex data science metrics interpretable for non-technical administrators, the deployment integrates Google Generative AI (Gemini 2.5 Flash) interfaces, supplying the model with structured data context (district predictions, optimizer outputs, and national trends) to generate grounded natural language policy summaries.

VIII. LIMITATIONS AND FUTURE WORK

While SchemeImpactNet delivers high performance and measurable optimization gains, it exhibits structural constraints that present opportunities for future research:

- **Single Data Source:** The implemented model relies exclusively on the MoRD MGNREGA MIS portal. Integration of external signals such as rainfall anomalies from IMD or poverty indices from NITI Aayog could improve predictive accuracy for climate-sensitive districts, but requires consistent district-year granularity that was unavailable for the current study period.
- **Constant Returns-to-Scale Assumption:** The LP engine assumes a fixed linear relationship between budget and person-days (constant efficiency). This omits diminishing returns caused by administrative capacity ceilings and contractor bottlenecks at the district level, meaning the 6.38% gain is an upper-bound simulation rather than a guaranteed operational outcome.
- **Training-Serving Feature Skew:** The live inference endpoint approximates `state_lag1_zscore` using district-level history as a proxy rather than the state-aggregate computation used during training. Saving fitted transformers in the model bundle would eliminate this discrepancy.
- **Lack of Real-Time Pipelines:** The current pipeline uses batch-loaded data files, requiring manual updates rather than direct synchronization with live government MIS portals.

Future research will focus on replacing the static linear programming configuration with a multi-period optimization agent capable of handling administrative capacity constraints and non-linear returns-to-scale. Additionally, incorporation of satellite-derived vegetation and drought indices may improve predictive accuracy within ecologically vulnerable districts.

IX. CONCLUSION

This paper presented SchemeImpactNet, a reproducible data science framework developed to transform public welfare administration from descriptive historical management into proactive, evidence-based decision support. By organizing and evaluating a panel dataset of 6,230 district-year observations within a strict walk-forward temporal cross-validation protocol, we demonstrate that non-parametric Gradient Boosting configurations achieve a robust mean R^2 of 0.9078 (excluding the 2022 West Bengal administrative reporting anomaly), outperforming classic parametric linear formulations. A rigorous leakage audit reduced an inflated initial R^2 of 0.9963 to this

honest estimate by removing target-correlated proxy variables. By connecting these continuous localized forecasts with a two-stage linear programming optimization engine implemented in SciPy, the framework establishes that mathematically optimized resource reallocation can expand nationwide welfare delivery by 1,840.41 lakh person-days — a 6.38% net efficiency gain achieved without increasing nominal public spending, under constant returns-to-scale assumptions. The containerized web deployment, augmented with Gemini large language model interpretation layers, confirms that complex data analytics can be effectively translated into accessible tools for public administrators, advancing the objectives of transparent, data-driven governance.

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